

Scalability & Future Plan

Date	30 September 2026
Team ID	SWTID-2026-3326
Project Name	SWTID-2026-3326
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Current system is mainly designed for local/small-scale usage	performance may decrease with many simultaneous users	High
2	MongoDB storage may require optimization as blog data increases	Larger datasets can increase query and storage requirements	Medium
3	AI blog generation depends on the external Google Gemini API	API availability and response time can affect AI generation	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for limited users	Deploy on scalable cloud infrastructure and use load balancing
2	Data Storage	MongoDB database	Use optimized indexes, database scaling, and backup strategies
3	Performance	Node.js and Express.js API	Add caching, API optimization, and horizontal scaling
4	Security	User authentication and API-based access	Implement stronger authentication, rate limiting, validation, and secure environment configuration

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Advanced search, categories,	1–2 Months	Better content discovery and user
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	and improved blog management		experience
Phase 3	AI-powered blog summarization, recommendations, and content enhancement	3–4 Months	Improved AI-assisted blogging capabilities
Phase 4	Cloud deployment, advanced security, analytics, and scalable infrastructure	5–6 Months	Support larger user base and production-level usage